

1. Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait (). 5. Display the Process ID (PID) of both parent and child processes.

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>
#include <string.h>

int main()
{
    char cmd[100];

    printf("Enter Linux Command: ");
    scanf("%99s", cmd);

    pid_t pid = fork();

    if (pid < 0)
    {
        perror("Fork Failed");
        return 1;
    }
    else if (pid == 0)
    {
        printf("\n----- Child Process ----- \n");
        printf("Child PID : %d\n", getpid());
        printf("Parent PID : %d\n", getppid());

        execlp(cmd, cmd, NULL);

        perror("Command Execution Failed");
        exit(1);
    }
    else
    {
        printf("\n----- Parent Process ----- \n");
        printf("Parent PID : %d\n", getpid());
        printf("Child PID : %d\n", pid);

        wait(NULL);

        printf("\nChild Process Completed.\n");
    }

    return 0;
}
```

```
bash: ./: is a directory
adithya@LAPTOP-07A78KDI:~/OS/Skill1$ ./skill1
Enter Linux Command: ls -l
./skill1
----- Parent Process -----
Parent PID : 2494
Child PID : 2499

----- Child Process -----
Child PID : 2499
Parent PID : 2494
skill1 skill1.c

Child Process Completed.
```

Using Linux terminal commands (uname, lscpu, lshw, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

```
adithya@LAPTOP-07A78KDI:~/OS$ uname -a
Linux LAPTOP-07A78KDI 6.18.33.1-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC
Fri Jun  5 01:12:21 UTC 2026 x86_64 GNU/Linux
```

```
adithya@LAPTOP-07A78KDI:~/OS$ lscpu
Architecture:                x86_64
CPU op-mode(s):              32-bit, 64-bit
Address sizes:               39 bits physical, 48 bits virtual
Byte Order:                  Little Endian
CPU(s):                      16
On-line CPU(s) list:         0-15
Vendor ID:                   GenuineIntel
Model name:                  13th Gen Intel(R) Core(TM) i7-13620H
CPU family:                  6
Model:                      186
Thread(s) per core:          2
Core(s) per socket:          8
Socket(s):                   1
Stepping:                    2
BogoMIPS:                    5836.80
Flags:                       fpu vme de pse tsc msr pae mce cx8 apic sep mtr
                             r pge mca cmov pat pse36 clflush mmx fxsr sse s
                             se2 ss ht syscall nx pdpe1gb rdtscp lm constant
                             _tsc rep_good nopl xtopology tsc_reliable nonst
                             op_tsc cpuid tsc_known_freq pni pclmulqdq vmx s
                             sse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe p
                             opcnt tsc_deadline_timer aes xsave avx f16c rdr
                             and hypervisor lahf_lm abm 3dnowprefetch ssbd i
                             brs ibpb stibp ibrs_enhanced tpr_shadow ept vpi
                             d ept_ad fsgsbase tsc_adjust bml avx2 smep bmi
                             2 erms invpcid rdseed adx smap clflushopt clwb
                             sha_ni xsaveopt xsavec xgetbv1 xsaves avx_vnni
                             vnni umip waitpkg gfni vaes vpclmulqdq rdpid mo
                             vdiri movdir64b fsrm md_clear serialize ibt flu
                             sh_llld arch_capabilities

Virtualization features:
  Virtualization:            VT-x
  Hypervisor vendor:          Microsoft
  Virtualization type:        full

Caches (sum of all):
  L1d:                       384 KiB (8 instances)
  L1i:                       256 KiB (8 instances)
  L2:                         10 MiB (8 instances)
  L3:                         24 MiB (1 instance)

NUMA:
  NUMA node(s):              1
  NUMA node0 CPU(s):         0-15

Vulnerabilities:
  Gather data sampling:       Not affected
  Ghostwrite:                 Not affected
  Indirect target selection:   Not affected
  Itlb multihit:              Not affected
  L1tf:                       Not affected
  Mds:                        Not affected
  Meltdown:                   Not affected
  Mmio stale data:            Not affected
  Old microcode:              Not affected
  Reg file data sampling:      Mitigation; Clear Register File
  Retbleed:                   Mitigation; Enhanced IBRS
  Spec rstack overflow:        Not affected
  Spec store bypass:          Mitigation; Speculative Store Bypass disabled v
                               ia prctl
  Spectre v1:                 Mitigation; usercopy/swapgs barriers and __user
                               pointer sanitization
  Spectre v2:                 Mitigation; Enhanced / Automatic IBRS; IBPB con
                               ditional; PBRSE-eIBRS SW sequence; BHI BHI_DIS_
                               S
  Srbds:                      Not affected
  Tsa:                        Not affected
  Tsx async abort:            Not affected
  Vmscape:                    Not affected
```

```

Vmscape: Not at
adithya@LAPTOP-07A78KDI:~/OS$ ps
  PID TTY          TIME CMD
  324 pts/0        00:00:00 bash
  646 pts/0        00:00:00 ps
adithya@LAPTOP-07A78KDI:~/OS$ |

```

```

Tasks:  24 total,   1 running,  23 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,   0.0 sy,   0.0 ni,100.0 id,   0.0 wa,   0.0 hi,   0.0 si,   0.0 st
MiB Mem :  7755.6 total,  6530.4 free,   566.7 used,   810.8 buff/cache
MiB Swap:  2048.0 total,  2048.0 free,    0.0 used,   7188.9 avail Mem

```

| PID | USER     | PR | NI | VIRT   | RES   | SHR   | S | %CPU | %MEM | TIME+   | COMMAND         |
|-----|----------|----|----|--------|-------|-------|---|------|------|---------|-----------------|
| 1   | root     | 20 | 0  | 24188  | 15292 | 11488 | S | 0.0  | 0.2  | 0:00.55 | systemd         |
| 2   | root     | 20 | 0  | 3180   | 2204  | 2072  | S | 0.0  | 0.0  | 0:00.01 | init-systemd(Ub |
| 6   | root     | 20 | 0  | 3180   | 2048  | 1940  | S | 0.0  | 0.0  | 0:00.00 | init            |
| 47  | root     | 19 | -1 | 50428  | 16664 | 15380 | S | 0.0  | 0.2  | 0:00.13 | systemd-journal |
| 79  | systemd+ | 20 | 0  | 22416  | 14500 | 12004 | S | 0.0  | 0.2  | 0:00.06 | systemd-resolve |
| 86  | root     | 20 | 0  | 35412  | 12384 | 9252  | S | 0.0  | 0.2  | 0:00.47 | systemd-udevd   |
| 125 | root     | 20 | 0  | 2888   | 1948  | 1820  | S | 0.0  | 0.0  | 0:00.03 | chronyd-starter |
| 132 | root     | 20 | 0  | 4472   | 3092  | 2840  | S | 0.0  | 0.0  | 0:00.00 | cron            |
| 137 | message+ | 20 | 0  | 8792   | 5508  | 4828  | S | 0.0  | 0.1  | 0:00.04 | dbus-daemon     |
| 164 | root     | 20 | 0  | 44096  | 29688 | 14544 | S | 0.0  | 0.4  | 0:00.19 | networkd-dispat |
| 170 | root     | 20 | 0  | 18436  | 9404  | 8160  | S | 0.0  | 0.1  | 0:00.07 | systemd-logind  |
| 194 | syslog   | 20 | 0  | 220548 | 5812  | 4516  | S | 0.0  | 0.1  | 0:00.05 | rsyslogd        |
| 250 | _chrony  | 20 | 0  | 23732  | 11100 | 7116  | S | 0.0  | 0.1  | 0:00.06 | chronyd         |
| 260 | _chrony  | 20 | 0  | 12096  | 2452  | 1808  | S | 0.0  | 0.0  | 0:00.00 | chronyd         |
| 271 | root     | 20 | 0  | 123460 | 32712 | 18100 | S | 0.0  | 0.4  | 0:00.14 | unattended-upgr |
| 289 | root     | 20 | 0  | 5492   | 2724  | 2540  | S | 0.0  | 0.0  | 0:00.00 | agetty          |
| 319 | root     | 20 | 0  | 3196   | 1112  | 980   | S | 0.0  | 0.0  | 0:00.00 | SessionLeader   |
| 321 | root     | 20 | 0  | 3196   | 1248  | 1108  | S | 0.0  | 0.0  | 0:00.01 | Relay(324)      |
| 324 | adithya  | 20 | 0  | 6268   | 5524  | 3848  | S | 0.0  | 0.1  | 0:00.08 | bash            |
| 327 | root     | 20 | 0  | 8648   | 5636  | 4792  | S | 0.0  | 0.1  | 0:00.01 | login           |
| 381 | adithya  | 20 | 0  | 22344  | 13504 | 10888 | S | 0.0  | 0.2  | 0:00.13 | systemd         |
| 383 | adithya  | 20 | 0  | 22916  | 4100  | 2096  | S | 0.0  | 0.1  | 0:00.00 | (sd-pam)        |
| 411 | adithya  | 20 | 0  | 6136   | 5420  | 3864  | S | 0.0  | 0.1  | 0:00.08 | bash            |
| 660 | adithya  | 20 | 0  | 8208   | 5656  | 3520  | R | 0.0  | 0.1  | 0:00.00 | top             |